

Software Quality Assurance

Lecture 7

Outline

- Cost of Quality

Cost of quality

Cost of quality (COQ) is defined as a methodology that allows an organization to determine the extent to which its resources are used for activities that prevent poor quality, that appraise the quality of the organization's products or services, and that result from internal and external failures.

Quality-related activities that incur costs may be divided into prevention costs, appraisal costs, and internal and external failure costs

Cost of quality

COQ = Failure Cost, Appraisal Cost & Prevention Cost

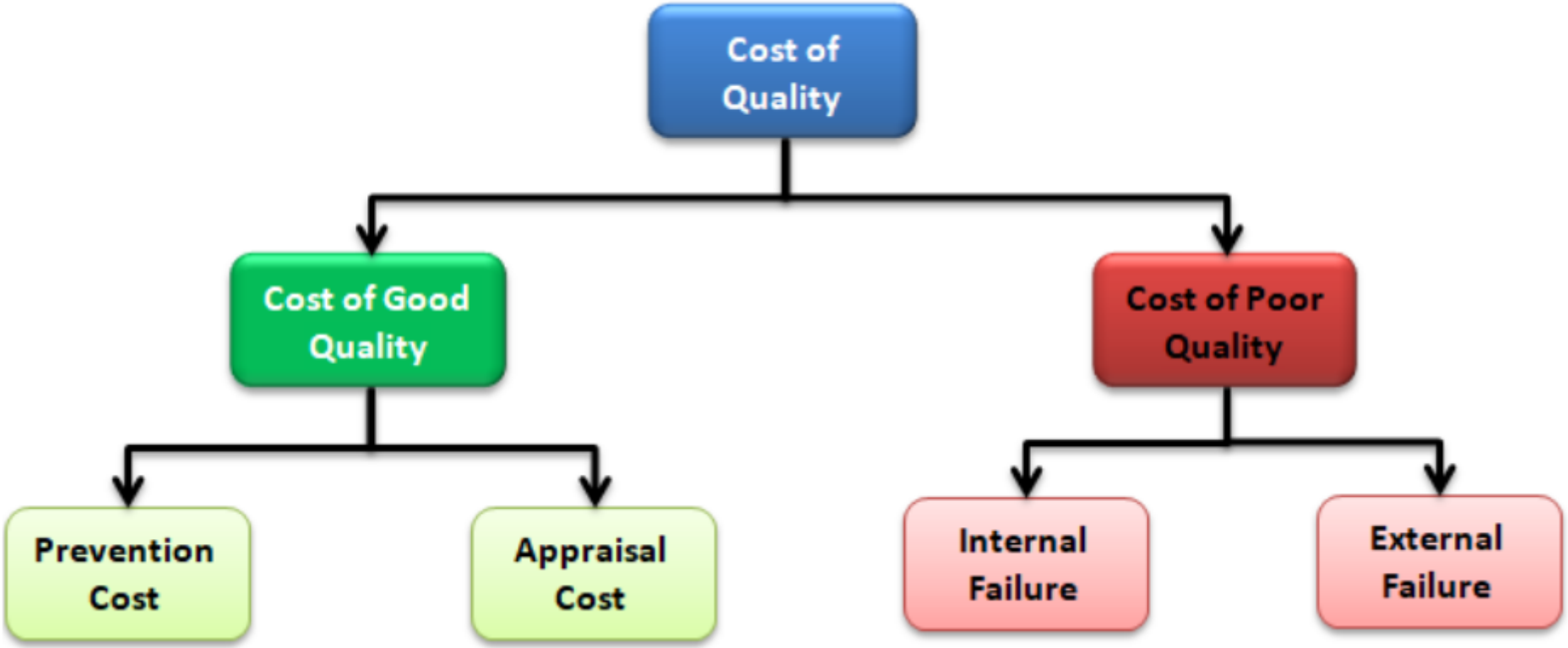
Failure Cost
- Cost of fixing defects
- running debugger to find defects
- repairing time, redesigning, recompiling or retesting

Appraisal Cost
- work to assess the product to see if it has defects
- excluding time spent on fixing
- does not include repair cost
- cost paid to assure that product is defect free

Prevention Cost
- cost of modifying the process
- ex. Analysis done to understand defects
- redesign and test of new process
- building prototype



Cost of Quality Example


Cost of Quality
Cost of Good Quality
Cost of Poor Quality
Prevention Cost
Appraisal Cost
Internal Failure
External Failure

Example

- **Cost of quality = CoGQ + CoPQ**
  - **CoGQ cost of good quality**
  - **CoPQ cost of poor quality**

Simplified Calculations

- Appraisal COQ.. Sum of all review time as % of Total Development Time
- Failure COQ.. Sum of all compile + Test time as % of Total Development Time